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Technology Drives Us

build and re-build homes till these perfectly match the requirements of the owner. Quite a few 3D printers are available now for architectural and construction applications. However, a machine revealed at Maker Faire held in October 2014 in Rome stands out.

The three-armed delta machine showcased by an Italian company, called World's Advanced Saving Project (WASP), can be carried as a kit and easily assembled on-site in just two hours. The huge, but portable, machine is held together mainly with heavy-duty ratchet straps. Once filled with native mud and fibre, the machine starts its home-making job.

Critics at Maker Faire lauded the machine, as it stood out from its predecessors that require custom building materials for 3D printing. Extruders of WASP machine, on the other hand, can work with native materials like clay, which makes them handier in building homes for the homeless in impoverished nations. Sometime in 2015, WASP might start its work in Sardinia, building homes of mud reinforced with wool. The machine can be easily carried from one place to another to build more homes of treated mud.

WASP is also exploring if the machine can be used to 3D-print bone implants with ceramics like hydroxylapatite, bio-glass and aluminium oxide. The company is constantly researching the use of native materials, and has apparently developed a clay filament with the precision and control of plastic materials. Their latest 3D printer is capable of printing clay using a 0.35mm nozzle, almost equalling the precision and control of commercial fused deposition modelling (FDM) based plastic extruding 3D printers.

Getting some help from the Sun

A machine that can build homes in remote villages sounds nice. But, isn't power a problem in such places? How can the machine have a steady

supply of current to do its job? Its developers can perhaps take a clue from another technology developed independently, elsewhere, by 3D-printing-thought-leader Joshua Pearce.

Pearce, who is well-versed in 3D printing and solar technologies, is known for developing open source, low-cost 3D printers. Recently, he added two more solutions, both open source and solar-powered, to his useful repertoire.

The first scheme comprises an array of solar photovoltaic panels and a stand-alone printer. It is quite cost-efficient and can make very good models, but is more appropriate for schools and businesses as it is not very portable. It can be stationed at a sunny yard and used effectively.

The second solar-powered solution is more portable. It uses a RepRap printer, which is small enough to fit in a suitcase but has a lesser capacity than the previous option. It can be taken by social workers to backward communities to make essential stuff needed by them. Since it is a RepRap, it can even be used to make another RepRap or print parts to make a larger 3D printer.

Quasi-crystals make a strong entry

As more and more industries, ranging from aerospace to healthcare, resort to 3D printing to manufacture equipment, it becomes important to have lighter materials with stronger functional properties. Researchers at University of Lorraine in France say that new composite materials developed with the help of quasi-crystals, a type of complex metal alloy (CMA) with crystal-like properties, can meet that demand.

CMAs have very desirable properties like low friction, corrosion-resistance and durability. However, being brittle, it is difficult to use them as the main bulk material. The team at Lorraine suggests that instead these can be used as